
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Mwesigye, Peter
Date of birth:	9/19/1978
Subject:	030-Flight Performance&Planing
Result:	35,69 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	375	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
2	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
3	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	382	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
6	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
7	388	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
8	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	387	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	380	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
12	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
13	376	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
14	183	31.2	LOADING	0,00	1,00
15	278	31.3.3	Basic calculations of CG	0,00	2,00
16	425	32.	PERFORMANCE - AEROPLANES	0,00	1,00
17	328	32.	PERFORMANCE - AEROPLANES	0,00	1,00
18	423	32.	PERFORMANCE - AEROPLANES	0,00	1,00
19	106	32.1.2.2	Definitions, Terms and Concepts	0,00	1,00
20	152	32.2.5.4	Landing	0,00	1,00
21	36	32.2.5.4	Landing	1,00	1,00
22	93	32.3.1	Definitions of terms and speeds	0,00	1,00
23	108	32.4.1	Take - off	1,00	1,00
24	101	32.4.1.7	Take-off climb	0,00	1,00
25	29	32.4.1.8	Obstacle-limited take-off	0,00	1,00
26	167	32.4.3	Cruise	0,00	1,00
27	47	32.4.3	Cruise	1,00	1,00
28	117	32.4.3.4	Long Range Cruise	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	196	32.4.5	Descent	0,00	1,00
30	21	32.4.5.1	Descent techniques	0,00	1,00
31	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
32	419	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
33	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	1,00	1,00
34	148	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
35	221	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	1,00	1,00
36	259	33.1.1.5	Completion of navigation plan	0,00	2,00
37	175	33.2.1.2	Courses and distances from en-route charts	0,00	1,00
38	252	33.4.1.3	Airway routings and airspace structure	1,00	1,00
39	88	33.5.1.1	Format of Flight Plan	1,00	1,00
40	30	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:36%				15,00	42,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Mwesigye, Peter

Location: Uganda Civil Aviation Authority

Your Result: 15,00 from 42,00 Points (35,71 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	6,00	16,00	37,50
31.2. LOADING	1	0,00	1,00	0,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	1	0,00	2,00	0,00
32. PERFORMANCE - AEROPLANES	15	4,00	15,00	26,67
32.1. GENERAL	1	0,00	1,00	0,00
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	2	1,00	2,00	50,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	1	0,00	1,00	0,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	8	3,00	8,00	37,50
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	5,00	11,00	45,45
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	5	2,00	6,00	33,33
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.4. PRE-FLIGHT PREPARATION	1	1,00	1,00	100,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	2	1,00	2,00	50,00
Total	40	15,00	42,00	35,71

1

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb
CG location Station 80.0

☐ 70.0 pounds.

☒ 35.9 pounds.

☐ 69.5 pounds.

2

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☒ 968.6 pounds.

☐ 924.5 pounds.

☐ 953.6 pounds.

3

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb

CG location Station 90.0

Aft CG limit Station 90.5

☐ **161.0 pounds.**

☒ **110.5 pounds.**

☐ **61.0 pounds.**

4

The center of gravity of an aircraft can be determined by

- ☐ dividing total moment by total weight.
- ☒ multiplying total arm by total weight.
- ☐ dividing total arm by total moments.

5

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.
- ☐ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.
- ☒ The aft CG provides the highest stall speed and cruising speed.

6

As the CG location is changed, recovery from a stall becomes progressively

- ☐ more difficult as the CG moves rearward.
- ☒ less difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.

7

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☐ There is no relationship between CG location and indicated airspeed.
- ☐ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.
- ☒ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.

Candidate: Peter Mwesigye
Date of birth: 1978-09-19

Subject: 030-Flight Performance&Planing
Date: 2014-07-18

8

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☒ **85.1.**

☐ **84.1.**

☐ **84.9.**

Candidate: Peter Mwesigye
Date of birth: 1978-09-19

Subject: 030-Flight Performance&Planing
Date: 2014-07-18

9

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 81.3.

☐ 80.6.

☒ 83.4.

10

As the CG moves aft, an aircraft becomes

- ☐ less stable, yet easier to control.
- ☒ more stable and controllable as long as the aft CG is not exceeded.
- ☐ less stable and less controllable.

11

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

☐ No; the new CG would be located at Station 116.89.

☐ Yes; the new CG would be located at Station 117.89.

☒ No; the new CG would be located at Station 116.42.

12

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☒ 62.5 pounds.

☐ 65.8 pounds.

☐ 68.9 pounds.

13

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☒ **78.1.**

☐ **79.9.**

☐ **77.9.**

14

In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:

- ☒ equal to half the weight of the aircraft multiplied by the semi span
- ☐ lowest at the wing root
- ☐ equal to the zero -fuel weight multiplied by the span
- ☐ highest at the wing root

Candidate: Peter Mwesigye
Date of birth: 1978-09-19

Subject: 030-Flight Performance&Planing
Date: 2014-07-18

15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

☐ $A = B + Fb / Fa$

☐ $A = B (Fa + Fb)$

☒ $A = B \times Fa / Fb$

☐ $A = B \times Fb / Fa$

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

16

During recoveries from unusual attitudes, level flight is attained the moment

- ☒ **a zero rate of climb is indicated on the VSI.**
- ☐ **the altimeter and airspeed needles stop prior to reversing their direction of movement.**
- ☐ **the horizon bar on the attitude indicator is exactly overlapped with the miniature airplane.**

17

(Refer to Figures 84 and 85.) What is the approximate fuel consumed when holding under Operating Conditions H-4?

☒ **3,190 pounds.**

☐ **2,550 pounds.**

☐ **3,050 pounds.**

See Attachments:

1. atp_84.jpg	figure 84	(Attachment No.2)
2. atp_85.jpg	figure 85	(Attachment No.3)

18

To level off at an airspeed higher than the descent speed, the addition of power should be made, assuming a 500 FPM rate of descent, at approximately

- ☒ 50 to 100 feet above the desired altitude.
- ☐ 150 to 200 feet above the desired altitude.
- ☐ 100 to 150 feet above the desired altitude.

19

Consider the graphic representation of the power required versus true air speed (TAS), for a jet aeroplane with a given mass. When drawing the tangent out of the origin, the point of contact determines the speed of:

- ☒ decreased
- ☐ very significantly decreased
- ☐ unchanged
- ☐ increased

20

Which of the following factors will lead to an increase of ground distance during a glide, while maintaining the appropriate minimum glide angle speed?

☐ Decrease of aircraft mass.

☒ Headwind.

☐ Increase of aircraft mass.

☐ Tailwind.

21

The coefficient of lift can be increased either by flap extension or by

- ☒ increasing the angle of attack.
- ☐ decreasing the 'nose-up' elevator trim setting.
- ☐ increasing the CAS.
- ☐ increasing the TAS.

22

The speed V_S is defined as

- ☐ safety speed for take-off in case of a contaminated runway.
- ☐ stalling speed or minimum steady flight speed at which the aeroplane is controllable.
- ☒ speed for best specific range.
- ☐ design stress speed.

23

The speed VR

☒ is the speed at which rotation to the lift-off angle of attack is initiated.

☐ must be equal to or lower than V1.

☐ must be higher than VLOF.

☐ must be higher than V2.

24

The climb gradient of an aircraft after take-off is 6% in standard atmosphere, no wind, at 0 ft pressure altitude. Using the following corrections: " $\pm 0,2\%$ / 1 000 ft field elevation", " $\pm 0,1\%$ / °C from standard temperature", " - 1 % with wing anti-ice" " - 0,5% with engine anti-ice", The climb gradient after take-off from an airport situated at 1 000 ft, 17° C; QNH 1013,25 hPa, with wing and engine anti-ice operating for a functional check is :

☒ 0.04

☐ 0.03

☐ 0.04

☐ 0.04

25

Which statement, in relation to the climb limited take-off mass of a jet aeroplane, is correct?

- ☒ The climb limited take-off mass is determined at the speed for best rate of climb.
- ☐ 50% of a head wind is taken into account when determining the climb limited take-off mass.
- ☐ The climb limited take-off mass decreases with increasing OAT.
- ☐ On high elevation airports equipped with long runways the aeroplane will always be climb limited.

26

Density altitude is the

- ☒ altitude reference to the standard datum plane
- ☐ altitude read directly from the altimeter
- ☐ pressure altitude corrected for 'non standard' temperature
- ☐ height above the surface

27

The optimum cruise altitude is

- ☐ the pressure altitude at which the speed for high speed buffet as TAS is a maximum.
- ☐ the pressure altitude at which the fuel flow is a maximum.
- ☐ the pressure altitude up to which a cabin altitude of 8000 ft can be maintained.
- ☒ the pressure altitude at which the best specific range can be achieved.

28

The optimum cruise altitude increases

- ☐ if the temperature (OAT) is increased.
- ☐ if the aeroplane mass is increased.
- ☒ if the aeroplane mass is decreased.
- ☐ if the tailwind component is decreased.

29

A jet aeroplane descends with constant Mach number. Which of the following speed limits is most likely to be exceeded first?

☐ **Maximum Operational Mach Number**

☒ **Never Exceed Speed**

☐ **High Speed Buffet Limit**

☐ **Maximum Operating Speed**

30

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ A tailwind component increases the ground distance.
- ☐ A tailwind component decreases the ground distance.
- ☐ A tailwind component increases fuel and time to descent.
- ☒ A headwind component increases the ground distance.

31

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.
- ☒ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.

32

All fuel tanks should be fired during preflight to determine

- ☐ if the pilot light functions properly on each tank.
- ☐ burner pressure and condition of the valves.
- ☒ if there are any leaks in the tanks.

33

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☒ **1 hour.**

☐ **2 hours.**

☐ **30 minutes.**

34

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☒ B AIP (Aeronautical Information Publication)

☐ D ATCC broadcasts

☐ A NOTAM

☐ C SIGMET

35

VFR flights shall not be flown over the congested areas of cities at a height less than

- ☐ C 500 ft above the heighest obstacle.
- ☐ B 2000 ft above the heighest obstacle within a radius of 600 ft from the aircraft.
- ☒ A 1000 ft above the heighest obstacle within a radius of 600 m from the aircraft.
- ☐ D the heighest obstacle.

36

Given: true course (TC) 017; W/V 340/30; TAS 420 kt
Find: wind correction angle (WCA) and ground speed (GS)

☒ WCA +2° ; GS 416 kt

☐ WCA -2° ; GS 396 kt

☐ WCA -2° ; GS 426 kt

☐ WCA +2° ; GS 396 kt

37

You are flying a constant compass heading of 252°. Variation is 22°E, deviation is 3°W and your INS is showing a drift of 9° right. True track is ?

☐ 280°

☒ 242°

☐ 262°

☐ 224°

38

From which of the following would you expect to find the dates and times when temporary danger areas are active

☒ **NOTAM and AIP**

☐ **SIGMET**

☐ **RAD/NAV charts**

☐ **Only AIP**

39

A "current flight plan" is a:

- ☐ *flight plan with the correct time of departure.*
- ☐ *filed flight plan.*
- ☒ *filed flight plan with amendments and clearance included.*
- ☐ *flight plan in the course of which radio communication should be practised between aeroplane and ATC.*

40

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☒ B XXXX

☐ D ZZZZ

☐ A AAAA

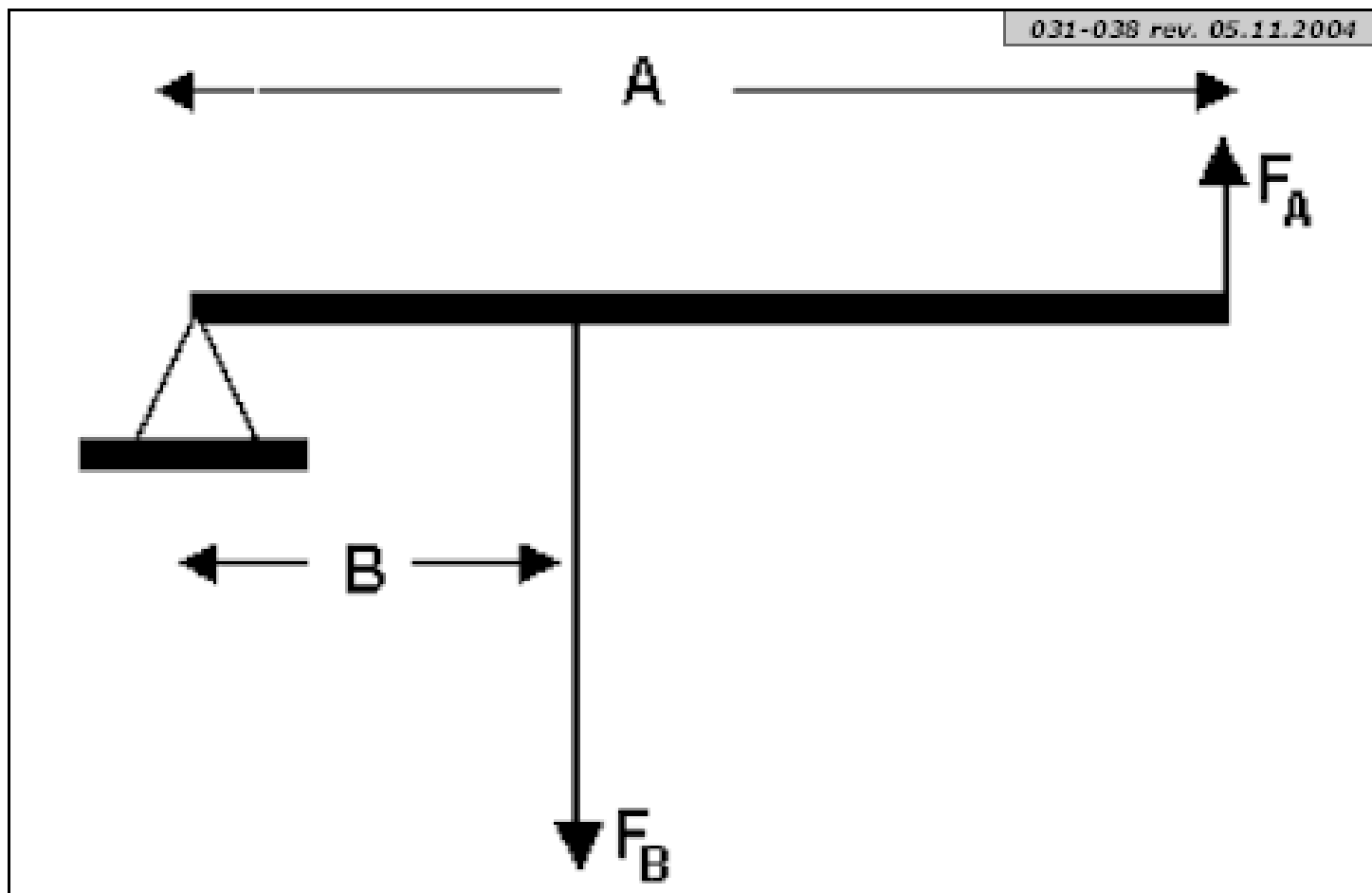
☐ C IIII

Attachment No. 1

Questions 15

Annex

031-038.gif



Attachment No. 2

Questions 17

figure 84

atp_84.jpg

OPERATING CONDITIONS	H-1	H-2	H-3	H-4	H-5
ALTITUDE	24,000	17,000	8,000	18,000	22,000
WEIGHT (X1000)	195	185	155	135	175
ENGINES OPERATING	3	3	3	3	3
HOLDING TIME (MIN)	15	30	45	25	35

Figure 84. B-727 – Holding © ASA

Attachment No. 3

Questions 17

figure 85

atp_85.jpg

EPR IAS - KTS FF PER ENG - LB/HR			HOLDING							B-727	
PRESSURE ALTITUDE FT	GROSS WEIGHT - 1000 LB										
	200	190	180	170	160	150	140	130	120		
25000	1.85 268 3600	1.81 261 3400	1.77 253 3210	1.73 246 3030	1.69 238 2860	1.64 230 2680	1.60 222 2510	1.55 213 2340	1.51 205 2180		
20000	1.69 265 3630	1.66 258 3450	1.62 251 3280	1.59 244 3110	1.55 236 2940	1.51 228 2770	1.48 220 2600	1.44 212 2440	1.40 204 2270		
15000	1.56 263 3670	1.53 256 3500	1.50 249 3340	1.47 242 3170	1.44 235 3000	1.41 227 2850	1.38 219 2680	1.35 211 2520	1.32 203 2350		
10000	1.45 262 3800	1.43 255 3640	1.40 248 3460	1.38 241 3310	1.35 234 3140	1.33 226 2970	1.30 218 2810	1.28 210 2640	1.25 202 2480		
5000	1.36 260 3890	1.34 254 3720	1.32 247 3550	1.30 240 3380	1.28 233 3220	1.26 225 3060	1.24 218 2890	1.22 210 2730	1.20 201 2560		

Figure 85. B-727 – Holding Performance Chart © ASA